25901585 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZGYA01 - AN UNDERSTANDING OF RIVERINE ECOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Choose the Correct Option

Answer all questions

Each question carries **1** marks

1. A continuous record of discharge plotted against time is called [U] / [CO1]

   a). Stream flow b). Flow pulse

   c). Hydrograph d). Flow regime

2. In which course of river are potholes most typically found ? [U] / [CO1, CO2, CO3, CO4, CO5]

   a). Lower Course b). Middle Course

   c). Upper Course d). Delta Course

3. The agricultural runoff leads to [A] / [CO2]

   a). Eutrophication b). Oxygen depletion in water

   c). Decrease in aquatic organisms d). All of the above

4. Stocking density means [U] / [CO3]

a). Total fish weight b). Number of fish per unit area

c). Amount of feed d). Water depth

5. The best time to visit Malarickal Village is during [U] / [CO5]

a). June - September b). January - June

c). October - March d). April - July

6. Which indicator best reflects biological State of a river? [U] / [CO2]

a). Temperature trend b). Species diversity

c). Urban growth rate d). Industrial output

7. Climate change affects rivers through _____. [U] / [CO2]

a). Stable rainfall b). Altered hydrology

c). Reduced erosion d). Increased oxygen

8. Which practice helps improve river water quality? [K] / [CO3]

a). Dumping waste b). Wastewater treatment

c). Channel narrowing d). Over-extraction of water

9. Which type of water body is most suitable for river cage culture? [U] / [CO3]

a). Stagnant water with no flow b). Highly polluted rivers

c). Moderately flowing clean rivers d). Dry river beds

10. Which of these is not a tributary of river Periyar? [K] / [CO4]

a). Muthirapuzha b). Edamalayar

c). Kallar d). Mangalapuzha

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. Discuss how human activities causes hydrologic alteration in rivers.	[U] / [CO1]
12. What is environmental flow?	[U] / [CO2]
13. Mention any two regulating services of rivers.	[K] / [CO3]
14. What is meant by River Bifurcation ?	[U] / [CO1, CO2, CO3, CO4, CO5]
15. What are riverine ecosystem services? Give one example.	[U] / [CO2]
16. Differentiate the catchment area and river basin.	[U] / [CO1, CO2, CO3, CO4, CO5]
17. Define system state in the DPSIR framework in relation to river ecosystems.	[K] / [CO2]
18. What is meant by a regional or local river restoration model?	[K] / [CO3]
19. What are allochthonous inputs ?	[U] / [CO1]
20. What are the main pollution abatement works taken up by National River Conservation Plan?	[U] / [CO4]
21. Give an accounts on aquatic larvae and adult insects in a riverine ecosystem.	[U] / [CO1]
22. List any two effects of dams on streamflow.	[U] / [CO2]
23. Why is ornamental fish farming popular?	[U] / [CO3]
24. What is Dissolved Oxygen (DO) and why is it important?	[U] / [CO1, CO2, CO3, CO4, CO5]
25. Mention any two major constraints in cage culture.	[An] / [CO3]
26. What are invertivores?	[U] / [CO1]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27.	Describe the Meenachil River MRRM model and its objectives.	[U] / [CO1, CO2, CO3, CO4, CO5]
28.	Describe the Convention on Biological Diversity (CBD) and its objectives.	[U] / [CO3]
29.	Classify mineral substrates in the riverbed based on partied size.	[U] / [CO1, CO2, CO3, CO4, CO5]
30.	Describe the major physical parameters used in river assessment.	[U] / [CO1, CO2, CO3, CO4, CO5]
31.	Describe Strahler stream order method.	[U] / [CO1, CO2, CO3, CO4, CO5]
32.	Discuss the concept of tourism carrying capacity and its importance in river management.	[K] / [CO3]
33.	Explain how riverfront development measures can be applied to reduce pollution and improve water quality in an urban river.	[A] / [CO3]
34.	Describe the Pen culture practices carried out in Kerala state.	[U] / [CO5]
35.	Explain the methods used for microbiological analysis of river water.	[U] / [CO1, CO2, CO3, CO4, CO5]
36.	What is longitudinal profile of a river ? Give its key characteristics.	[U] / [CO1, CO2, CO3, CO4, CO5]

[4x8 = 32]